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Coral morphology detection in underwater imagery using YOLOv12 with CNN and transformer encoder fusion

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For monitoring marine biodiversity and observing the health of reefs, it's important to be able to accurately see coral morphology in underwater images. Present approaches for finding underwater objects still have problems because light can be absorbed, scattered, or become turbid, and coral morphologies look very similar to each other, which often results in misclassification, erratic localization, and lower accuracy. The present automated detectors, like YOLOv7 and latest newer variants, like YOLOv11, do not work well with underwater image distortions, do not reason well about global features, or are not efficient enough to be used in practical marine monitoring systems. This paper introduces an innovative method for detecting coral morphology automatically from underwater images, utilizing YOLOv12 augmented with a Convolutional Neural Network (CNN) and transformer encoder integration in real-time to mitigate these constraints. The suggested model uses the local feature extraction power of CNN and the global context modelling power of transformers to improve feature representation in intricate underwater situations. The model is more accurate and efficient because it uses the strength of YOLOv12 along with the strengths of CNN and transformer encoders to detect objects. Empirical assessments on a standard underwater dataset show that our method is better than the present models in terms of precision, recall, and mean Average Precision (mAP), while still being able to make inferences in real time. This work helps make coral reef monitoring tools better by giving them a reliable and scalable way to identify objects underwater.

Keywords Deep learning, Coral reef ecosystems, Coral morphology detection, YOLOv12, CNN, Transformer, Fusion, Marine ecosystem monitoring, Real-time detection

Coral reefs are recognized as highly dynamic aquatic ecological community that preserve a wide variety of marine organisms. They protect the coast, aid a wide range of plant and animal life, and contributes to major environmental and financial services like fishing and tourism. But coral reefs are getting vitiated because of the changes in climate, acidification of the ocean, pollution, and anthropogenic activities. This makes it more vital than ever to keep an eye on coral health and shape. Biologically and functionally established coral growth patterns, like branching, massive, and tabular morphologies are defined on the basis of skeletal architecture, ecological strategies, and evolutionary adaptations[1–3]. Coral morphology detection has many uses in ecology and conservation employment because the way coral grows is intently related to the intricacy of the reef structure, the number of different species, and the resilience of the ecosystems[4–7]. Identifying morphologies automatically allows for the quantitative assessment of habitat quality, the prompt identification of stress induced by climate change, and the evaluation of long-term changes in reef community structure. These observations are especially useful for planning conservation efforts, like choosing morphologically varied reef regions to protect, guide restoration work by choosing morphological types which make habitats more complex, and helping models of fisheries productivity and coastal protection. Marine biologists manually annotate underwater images as part of conventional coral evaluation techniques. These methods are time-consuming, labour-intensive, and open to subjective interpretation. Because of this, there is an increasing demand of automatically and accurately recognition systems that can work well in intricate underwater settings. Automatic image analysis approaches can tell different coral morphologies apart by looking at them[8,9]. By adding morphology detection to autonomous

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survey platforms, researchers and managers can monitor reefs in real time and on a large scale. This makes it possible to manage marine ecosystems based on data in the face of rapid environmental transformation.

Few studies have underscored the efficacy of fusion-based methodologies for underwater and remote sensing image analysis[10–12]. Recent advancements in deep learning, specifically in object detection, has yielded encouraging outcomes for marine applications, such as fish tracking, habitat mapping, and coral identification[13–16]. The YOLO (You Only Look Once) models are now very good at detecting things in real time[17–20]. However, conventional detectors have trouble with underwater images because colour gets changed, contrast is lowered, and light is blocked in water, which makes it hard to get detailed spatial and contextual information. This paper presents a new hybrid framework that combines YOLOv12 with a module that fuses Convolutional Neural Networks (CNNs) and transformer encoders in order to get around these problems. CNNs are good at modelling local spatial features, while transformers are good at capturing long-range dependencies and the overall context. In the proposed method, we use the power of both models to make feature representation better and make detection faster in complicated underwater settings. However, our study is unique because it proposes a hierarchical cross-attention integration module that combines local texture representations from the CNN with global contextual relationships extracted by the transformer encoder whereas other models only combine CNN and transformer features. This module functions at multiple scales, and is fused right into the YOLOv12 feature pyramid to make it easier to detect objects in underwater images with low visibility. The work's main contributions are:

- We develop a hybrid detection method that uses CNN and transformer encoders in the YOLOv12 framework to make it easier to detect coral morphology.
- We create a feature fusion method that uses both local and global information present in underwater images with different lighting conditions.
- We test the suggested method on a standard dataset of underwater images and show that it is better than current detection models in terms of accuracy, precision, recall, and speed of inference.

The goal of this work is to provide a method to automatically monitor corals in an effective and accurate way to find coral morphology in practical underwater situations.

This paper is organized like this. Section 2 looks at other research on coral morphology detection, reef monitoring, and the latest improvements in deep learning architectures for ecological uses. In Sect. 3, we talk about the framework of the proposed method. We go into detail about the CNN and transformer encoder feature fusion strategy and how it works with YOLOv12. Section 4 talks about how the experiment was set up. Section 5 shows the results of the experiments. Lastly, Sect. 6 wraps up the paper and talks about where future research on automated, scalable reef monitoring should go.

Review of Literature

Current progress in deep learning has resulted in a variety of monitoring strategies, which can be generally classified into unsupervised, semi-supervised, supervised, and multimodal approaches. Supervised and multimodal frameworks have demonstrated robust performance over applications, ranging from underwater image enhancement, depth estimation, and reinforcement learning-based detection and tracking[21–27]. In the last few years, deep learning methods to automatically monitor coral reefs have become very popular, thanks to improvements in object-detection and classification architectures[28–30]. This section looks at the most important parts of the current body of literature and what it lacks.

In the past, conventional machine learning methods like Support Vector Machines (SVMs) depended upon hand-made features like colour and texture descriptors. Beijbom et al.[31] employed texture and color descriptors at multiple scales to classify coral reefs, attaining a modest efficiency level but encountering difficulties in generalizing to different underwater conditions, such as scattering of light, turbidity, and colour attenuation. These techniques were constrained by their dependency on manually created features and insufficient ability to capture the intricate variation that is common in underwater settings.

The use of CNNs has improved the work of classifying images, like corals. CNNs have shown a remarkable ability to learn complex spatial patterns and extract hierarchical features from raw image data. González-Rivero et al.[32] created a CNN-based model using the ResNet architecture to sort coral genera in a big set of reef images. They reported noteworthy improvements in accuracy compared to traditional approaches. More recently, Artates et al.[33] employed the DenseNet and Inception-v3 architectures to classify levels of coral bleaching. Zhuang et al.[34] suggested UWNet, a lightweight framework integrating CNNs with a Structured Space Model (SSM) for detecting small targets, reporting good detection accuracy for organisms (e.g., starfish and scallops), but with limited scope for coral morphology. In chaotic underwater images, CNN-based models are frequently restricted by local receptive fields, which make it challenging to model long-range dependencies. Moreover, their effectiveness deteriorates under different lighting and low-contrast conditions, which are frequent in underwater imagery[35–38].

The YOLO series of models has become popular because they offer a good mix of detection speed and accuracy. They are beneficial for underwater applications with limited resources, such as onboard Autonomous Underwater Vehicles (AUVs). Several papers demonstrate the use of YOLO variants to generate end-to-end monitoring systems that can perform near-real-time detection and consistent recognition of colonies in underwater videos[39]. Many marine applications, such as detecting fishes, identifying trash underwater, and keeping an eye on coral reefs, have used YOLOv3 and YOLOv4[40–42]. Furthermore, YOLOv3 and YOLOv4 have been effectively employed for differentiating coral species and identifying bleaching occurrences, demonstrating robustness against underwater noise and fluctuations in lighting[43,44]. For example, Gorro et al. applied YOLOv3 to detect coral colonies in fluorescence images and videos and indicated accepted level of the detection accuracy[43]. More recent studies have adopted YOLOv5 and YOLOv7 to classify complex coral morphologies because they can better extract features and learn from them[45–47]. Younes et al.[48] created an automatic coral detection system using YOLOv5 on a dataset of 580 underwater images. These images were made from 400 original samples utilizing augmentation. Their system was able to find corals quickly, but it had trouble in low-contrast situations and with a small number of species. A common version of YOLOv8 has been modified for underwater tasks by integrating features like attention mechanisms and lightweight convolutions. The CEH-YOLO model, which is a version of YOLOv8, uses a High-order Deformable Attention (HDA) module and an Enhanced Spatial Pyramid Pooling-Fast (ESPPF) module to find small objects better in low-contrast underwater images[49]. Lu et al. conducted another study that presented SCoralDet, a YOLO-based model featuring a Multi-Path Fusion Block (MPFB) and lightweight components such as GSConv. It attained a mAP50 of 81.9 for soft coral detection, overcoming difficulties associated with intricate coral structures and dense growth[50]. Gorro et al. investigated the identification of corals, seagrass, and seaweeds utilizing YOLOv9[51]. Shamsuddin et al. conducted a comparative analysis of YOLOv7, YOLOv8, and YOLOv9 for detecting coral reef fish below water[52]. Current improvements to YOLOv12 have made recognition even better with the addition of residual ELAN blocks and area attention procedures for use in underwater settings. One research has indicated that YOLOv12 greatly improves the accuracy of recognizing objects in murky waters[53]. The main goal of the study, though, was to find marine litter. In general, YOLO-based methods have shown promise as a way to monitor coral reefs on a large scale. They strike a good equilibrium among speed, accuracy, and the ability to work in difficult underwater settings.

In the field of computer vision, combining CNNs with transformer modules is becoming more popular because such combination can extract local features while also having comprehension of the complete image. This combination has been looked into in underwater images to solve problems such as chromatic distortion and poor visibility. One example is ViT-ClarityNet, which combines Vision Transformers (ViTs) and CNNs to improve underwater images. It uses skip connections to retain high-frequency details that would otherwise be missed because of light dispersion[54]. It concentrates on improving images, instead of directly identifying objects. Liu et al. proposed a CNN-transformer combination for detecting underwater objects, demonstrating improved generalization in varying illumination conditions[55]. Yang et al. also suggested a dual-stream encoder that combines residual CNNs with Swin transformers for underwater image enhancement to make images more accurate and reliable when they are not in good condition[56]. These kinds of hybrid models look like a good way to find coral morphology, specifically when used with fast object detectors like the YOLO family.

Even though there has been a lot of development, there are still some problems with coral morphology detection. Even though CNNs and YOLO-based models work well, they often have trouble generalizing features in underwater scenes that vary a lot. Additionally, current YOLO-based models, despite their efficiency, encounter difficulties with morphologically analogous species (e.g., branching versus tabular corals) in low-visibility environments, resulting in classification inaccuracies. Nonetheless, powerful in context modelling, transformer-based models are computationally intensive, hindering deployment on resource-constrained underwater platforms. Lastly, most research centres on overall object detection or image improvement, with little to no focus on fine-grained coral morphology classification, which is vital to comprehending the health of reefs and climate change-induced growth patterns.

Thus, there remains a research gap in integrating the capabilities of transformers and CNNs in the YOLOv12 to attain high detection accuracy as well as real-time inference. The current paper bridges the gap by suggesting a fusion of the CNN-transformer encoder into YOLOv12 for sturdy, scalable, and effective coral morphology identification.

Proposed methodology

This research introduces a new coral morphology detection system combining the potentials of CNN[57–59] and transformer models[60–62] into the YOLOv12 object detection method. The system is constructed to overcome the inherent issues of underwater images, such as low illumination, insufficient contrast, colour degradation, and occlusions, which tend to hamper the performance of typical object detection techniques. The proposed system design comprises three major parts: the backbone, the transformer encoder fusion module in the neck, and the detection head. The hybrid CNN and transformer encoder enhances the capacity of the system to distinguish different coral morphologies through the local and global extraction of distinct features. The overview of the architecture is displayed in Fig. 1.

The architecture starts with a pre-processing phase. All the images of corals are resized to a resolution of 640 × 640 pixels to fulfil the input requirements of YOLOv12. Pixel intensities are normalized in the range [0,1] for consistent representation of values across coral image datasets. Data augmentation is implemented for improved generalization across coral images with a combination of geometric transformations (rotation, horizontal flip, scaling, and cropping) and photometric adjustments (brightness, contrast, and hue jittering). YOLO-specific data augmentation strategies, such as Mosaic[63] and MixUp[64], are implemented for generating synthetic multi-object images, which enables the model to better recognize coral clusters as well as overlapping morphologies. Data Augmentation strategies, such as mosaic, enhances generalizability of the model for distinct underwater environments. This augmentation also benefits the transformer encoder because it uses different context patterns to understand long-distance dependencies in coral structures. In addition, these pre-processing steps contribute significantly to image normalisation, thus playing a key role in the better extraction of features in the following layers. After pre-processing, backbone takes as input, an underwater coral image of size $(B \times 3 \times H \times W)$, where B represents the batch size, 3 corresponds to the colour channel of the input image, and $H \times W$ denotes the spatial resolution of the image. To gradually extract discriminative visual information while scaling down spatial resolution, a series of convolutional layers, C3k2 blocks, and A2C2f modules are used.

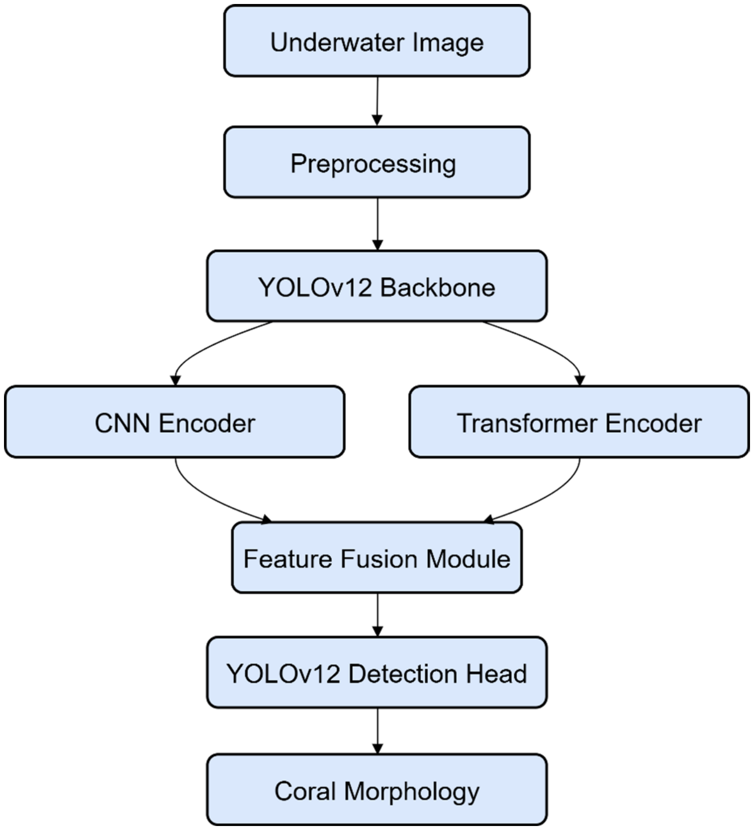

Fig. 1. Overview of the architecture of the proposed model.

Hierarchical feature maps at different scales are extracted by the backbone to capture low-level textural cues and high-level semantic patterns.

A CNN-transformer encoder fusion module is incorporated in the neck of YOLOv12 to improve feature expressiveness for detecting coral morphology. The feature maps are concurrently processed through two parallel feature encoding pathways comprising a CNN and a transformer encoder. The major innovation of the proposed approach is the hybrid neck design, that jointly integrates the feature aggregation capabilities of CNNs and the use of a transformer encoder module to improve global feature learning. The CNN branch concentrates on local feature networks, and it is especially good at getting high-resolution characteristics about coral morphology, like irregular textures and polyp features. The CNN uses a modified ResNet-50 architecture to extort spatial features like textures, edges, and structural patterns that are important for classifying coral species. At the same time, a transformer encoder module, utilizing a Swin transformer block[65], is employed to extract global context features. The highest-resolution feature map from the CNN backbone is given as input to a transformer encoder module. The transformer encoder block utilized is designed using the A2C2f framework and is comprised of layered attention blocks (A-blocks) and a convolutional projection layer. Area attention and optimized feed-forward layers are utilized in the transformer branch to model long-range dependencies across the scene to contextualize coral structures in relation to surrounding reef environments. The outputs of CNN and transformer encoders are merged through a specific feature fusion module, as shown. The module initially integrates features gathered from both streams, followed by channel and spatial attention mechanisms. A learnable scaling parameter, Self. gamma, is introduced in this module to adaptively regulate the contribution of transformer-derived features with respect to the original CNN features. Squeeze-and-Excitation (SE)[66,67] blocks are primarily used to recalibrate the channel, emphasizing the utmost informative features. However, spatial attention is of help in the location and refinement of salient areas corresponding to coral structures. Such a combination efficaciously combines the local detail sensitivity of CNNs with the global comprehension of transformers, delivering a more discerning and stronger feature representation.

Area attention is applied in the neck specifically to enhance demarcation among coral morphologies, and background clutter like rock, algae, or suspended particles. Strong detection across the scales is ensured by this layout, which enables the system to identify small, fine-structured coral tips and bigger branching colonies. The CNN feature maps, together with the transformer-enhanced representation, serve as input to the neck to perform a succession of up-sampling, concatenation, convolution, and A2C2f refinement operations. High level semantic information is proliferated to the feature maps having high resolution via up-sampling. Features of the different stages of the backbone are combined in concatenation. After that, convolutional and A2C2f layers improve these combined representations by reducing noise and highlighting characteristics that help tell the difference between morphologies. The neck makes three scale-consistent feature maps, represented as N_3, N_4,

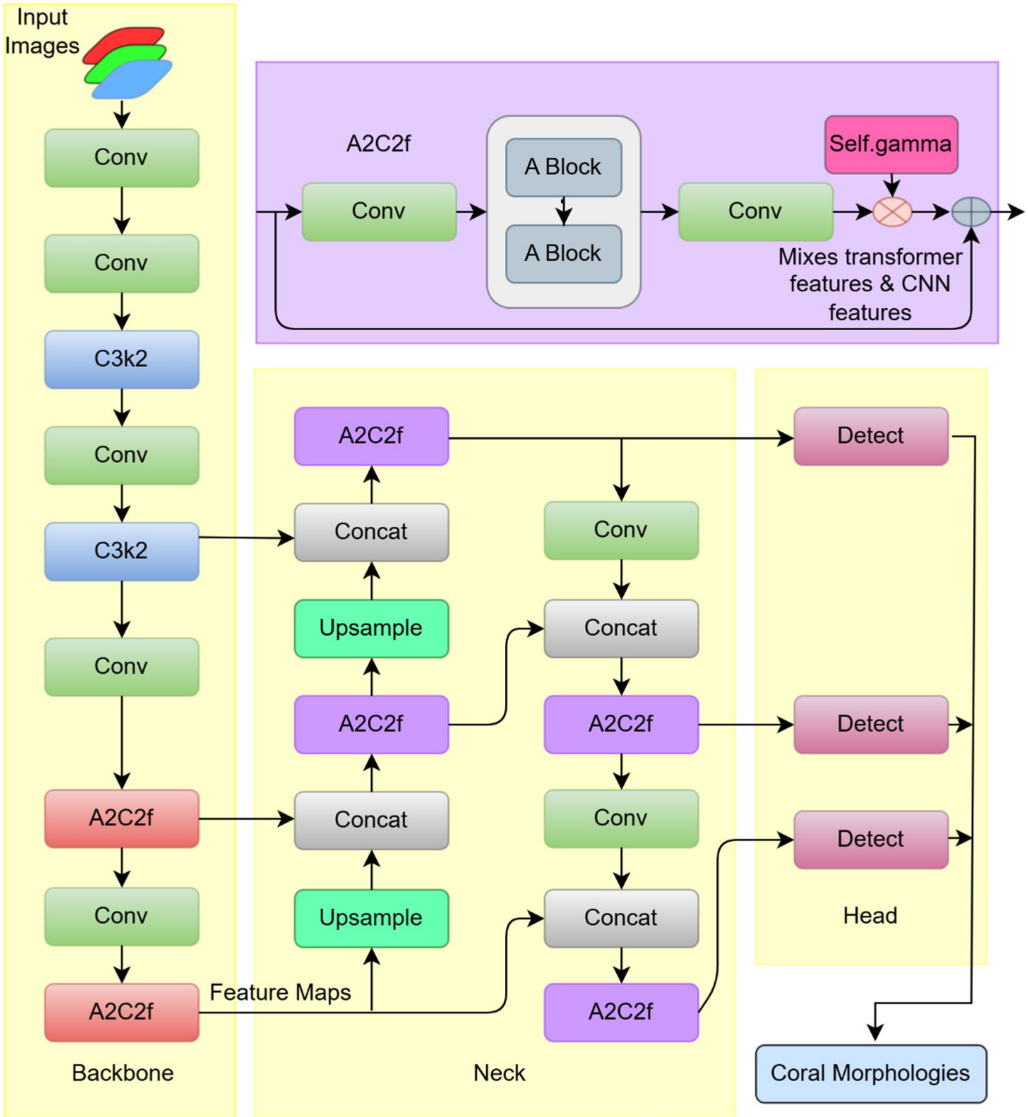

Fig. 2. Yolov12 architecture.

and N_5, for coral colonies that are small, medium, and large. There is a lot of size variation in coral reef images, and this multi-scale representation can show that variation well.

The YOLOv12 detection head then gets these combined features. It uses a parallel detection branch at each scale. Before making the actual detection, each prediction branch uses convolutional layers to make features even better. The head can pick up on small morphological differences that only convolutional detectors might miss by using these combined CNN-transformer features. Dynamic anchor generation, decoupled classification and regression components, and an improved Feature Pyramid Network (FPN) make the head work for multi-scale detection. The YOLOv12 detection head works as a single-stage detector and gives bounding boxes, confidence scores, and class predictions at different resolutions. Non-Maximum Suppression (NMS) is then used to get rid of duplicate predictions. Detection head sorts coral morphology into object classes based on morphological categories like arborescent, massive, and tabular. These changes to the model make it possible to find coral structures of all shapes and sizes quickly and more accurately.

Overall, the suggested methodology, intends to achieve a balance among efficiency and accuracy in detecting coral morphology through the collegial combination of CNNs and transformers in the framework of YOLOv12. The hybrid setup improves the capability to learn features while preserving real-time functionality, which makes this method very useful for monitoring of marine ecosystems.

Pseudocode

This segment explains the systematic pseudocode of the suggested YOLOv12-CNN-transformer combination technique for finding coral morphology. In steps 8 through 11, the backbone extracts features, followed by the local convolutional feature separation and improvement in the CNN refinement blocks. Steps 12 through 17 transform these CNN features into a series of tokens. The transformer encoder processes these features to get information about the whole context. Subsequently, these features are reconfigured back into spatial feature maps. Step 18 explains how to combine CNN-refined features with transformer-enhanced global features. The fused features are then passed into the YOLOv12 neck for multi-scale feature aggregation. Feature fusion is performed before the neck stage and after the backbone feature extraction, which is commonly referred to as neck-level (intermediate) fusion. Fusion at the neck stage balances semantic richness and spatial resolution. The detection head remains unchanged, preserving YOLOv12's efficient prediction mechanism.

Input:

Underwater coral image (I)

Output:

- Detected coral bounding boxes (B)
- Class labels (C) (coral morphology categories)
- Confidence scores (S)

```
1:  Initialize YOLOv12 backbone, neck, and detection head
2:  Initialize CNN feature refinement blocks
3:  Initialize Transformer Encoder with L layers and multi-head self-
attention
4:  Initialize feature fusion module

5:  procedure CORAL_DETECTION(I)
6:      I_pre ← Preprocess(I)
7:          resize image to fixed input resolution, normalize pixel values,
colour correction (optional)

8:      F_cnn ← YOLOv12_Backbone(I_pre)
9:          extract multi-scale convolutional feature maps

10:     F_ref ← CNN_Refinement(F_cnn)
11:         enhance local texture, edge features and fine-grained spatial
features

12:     F_seq ← Flatten_and_Embed(F_ref)
13:         convert spatial features to sequence of feature tokens

14:     F_trans ← Transformer_Encoder(F_seq)
15:         compute model global spatial dependencies

16:     F_trans_map ← Reshape(F_trans)
17:         restore spatial feature representation

18:     F_fused ← Feature_Fusion(F_ref, F_trans_map)
19:         concatenate or add CNN and Transformer features

20:     F_neck ← YOLOv12_Neck(F_fused)
21:         multi-scale feature aggregation

22:     P ← YOLOv12_Head(F_neck)
23:         predict bounding boxes, class scores, objectness

24:     B, C, S ← Postprocess(P)
25:         non-maximum suppression (NMS)

26:     return B, C, S
27: end procedure
```

Experimental setup

A series of controlled experiments were carried out to test the performance of the suggested YOLOv12 with a CNN-transformer encoder fusion model for coral morphology detection. This section describes the dataset, metrics for evaluation, training losses, and the implementation setting used in the research.

Dataset description

The experiments of this research use the coral dataset, which is part of the open Roboflow-100 (RF100)[68] benchmark group. The dataset was collected to advance object detection studies in marine and ecological settings, specifically for coral morphology classification. It contains high-resolution underwater images with well-defined coral growth forms. The dataset covers many significant morphological types, such as arborescent, tabular, and massive. These classes represent the various structural growth patterns commonly adopted in reef ecology for monitoring habitat complexity, biodiversity, and reef health. The dataset is mainly useful for benchmarking, as the images in the dataset were taken under strenuous settings with variable water turbidity, illumination variances, and occlusions. The conditions are similar to the difficulties found in in situ coral reef monitoring and serve as a good testbed for assessing the detection model. The dataset is divided into three disjoint groups: training, validation, and testing. The usual split ratio is used which is nearly 70%, 20%, and 10%, respectively. This work is in line with a well-known standard that facilitates replication and comparability with various cutting-edge detection models. Roboflow-100's coral dataset al.lows for evaluating an ecologically representative and meaningful set of underwater images, thus making the proposed detection model environmentally relevant.

Implementation environment

The proposed framework for coral morphology detection was tested using a high-performance computing setup capable of supporting the computational load required by deep learning models. The proposed coral morphology detection framework was implemented in a high-performance computing environment equipped to handle the computational demands of deep learning models. Experiments were executed on an NVIDIA A100 GPU with 40 GB RAM under CUDA acceleration. A multi-core CPU, Intel i7, was used. All images were resized to 640×640 pixels. SSD storage facilitated rapid access to a large underwater imagery dataset. The software environment was an Ubuntu 20.04 operating system, ensuring compatibility with GPU drivers and deep learning frameworks. The primary programming language used for implementation is Python 3.10, with the PyTorch 2.0 framework for deep learning. The HuggingFace Transformers library version 4.38.1 for transformer-based model integration, SciPy 1.12.0 and NumPy 1.26.4 for numerical computation, and OpenCV-Python 4.9.0.80 for image processing were additional dependencies. A batch size of 32 was used during both training and validation. The learning rate was set to 0.001.

Evaluation metrics

Universally adopted object detection metrics were used for evaluating the performance of the suggested model. The evaluation metrics give a thorough observation of detection accuracy, localization quality, and computational efficiency. Standard measures for performance in classification include accuracy, precision, recall, and F1-score[69,70]. For retrieval- and detection-oriented assessment, average precision and Mean Average Precision (mAP) are utilized because they are used to calculate precision-recall characteristics over decision thresholds[71,72]. For qualitative analysis of learned feature representations, t-SNE (t-Distributed Stochastic Neighbour Embedding) is employed for low-dimensional (typically 2D) visualization of high-dimensional embeddings[73] to assess how well the model splits distinct classes or clusters. Each point in the 2D t-SNE plot corresponds to the feature vector of a single coral image; pointing to the same morphology cluster together when the model has learned discriminative features from the data. Clusters that are clearly separated specify significant representations for various coral morphologies captured by the network, whereas overlapping clusters may indicate potential class confusions. Apart from predictive performance, the suggested model's computational efficiency and practical feasibility are measured by inference time[74].

Loss functions

The multiple loss components collectively optimize the training process of the suggested framework to accurately localize and classify the coral structures from underwater images. The total loss is defined as the sum of three main components: the loss due to bounding box regression, objectness, and classification. IoU-based loss is used to supervise bounding box regression loss, which is measured by calculating the geometrical overlap between the predicted and ground-truth boxes around the corals[75]. The bounding box regression loss reflects a discrepancy in the position, width, and height of the generated box with respect to the actual boundaries of the coral. Binary cross-entropy loss is used to optimize the objectness loss. This loss differentiates foreground objects from background regions. In contrast, the loss due to classification is usually cross-entropy or focal loss for addressing the class imbalance in densely packed detection tasks[76]. This classification loss assesses the model's ability to provide the correct class labels to detected corals and castigates misclassifying coral morphologies on the predicted boxes. The total loss is calculated by summing the weighted individual losses, hence balancing the trade-off between localization, confidence assessment, and category discernment.

$$L_{total} = \lambda_{box} L_{box} + \lambda_{cls} L_{cls} + \lambda_{obj} L_{obj} \tag{1}$$

where λ values are hyperparameters governing the comparative significance of each element, and their optimal values in the proposed work are $\lambda_{box} = 1.0$, $\lambda_{cls} = 0.5$, $\lambda_{obj} = 0.3$. These values were determined using the grid search optimization method. The combined loss function allows the model to simultaneously accurately localize coral regions (L_{box}), determine the presence of objects (L_{obj}), and correctly classify coral morphologies (L_{cls}), ensuring robust detection performance in complex underwater imagery.

Results and discussion

Figure 3 shows the loss convergence curves of the suggested YOLOv12 model integrating a CNN-transformer encoder fusion module. Every subplot reflects the development of the training loss: (a) bounding box regression loss, (b) objectness loss, (c) classification loss, and corresponding validation curves in (d), (e) and (f). In all six subplots, the blue trajectories show a decreasing trend without large fluctuations, which means the model progressively reduces detection and classification errors during training and demonstrates stable optimization and effective convergence. The gradual drop in bounding box regression loss shows that spatial localization was learned efficiently. The persistent drop in objectness loss reflects better discernment between coral structures and background clutter, which are characteristic of underwater images. In the same way, the progressive decadence in the classification loss indicates that the model distinguishes between coral morphological categories, such

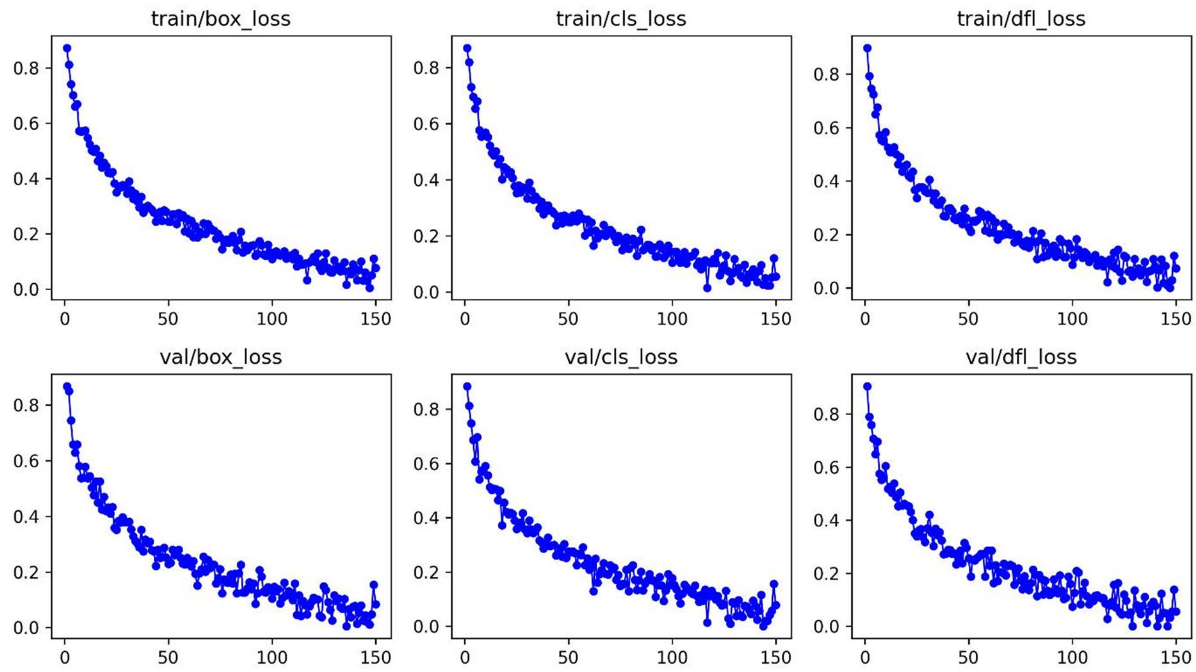

Fig. 3. Training and validation loss curves of the proposed framework.

as arborescent, massive, and tabular, to a greater extent. Steady convergence shown in this figure evidences the efficiency of the proposed CNN–transformer encoder fusion design in YOLOv12 for jointly optimizing localization, classification, and objectness, therefore demonstrating its applicability for precise coral morphology identification in actual underwater settings.

The sample detection outputs of the proposed YOLOv12 architecture incorporating a CNN–transformer encoder fusion are presented in Fig. 4, as applied to underwater coral reef images. Each image demonstrates a number of coral colonies under diverse visibility conditions with predicted bounding boxes and associated confidence scores for different morphological classes, such as arborescent, massive-poritidae, and tabular structures. The first row shows instances where the model correctly identifies regions devoid of corals, returning the label "No Coral Detected" with no spurious bounding boxes. This feature is very important for extensive reef observation as it minimizes false positives in frames obsessed with non-coral substrates or sand. The remaining rows demonstrate affirmative identifications, where various coral shapes are localized at the same time. Various morphological classes are represented by the differently coloured bounding boxes along with their corresponding confidence values. The results indicate that the model can process dense and morphologically varied reef images where many habitats of different growth patterns often overlay each other. For example, tabular corals are identified in a consistent manner with high levels of confidence, whereas massive-poritidae habitats are recognized even in conditions of partial obstruction or poor contrast. Some instances of coral recognition at lower confidence thresholds of approximately 0.3–0.4 are found, mostly related to arborescent morphology class. Based on these results, CNN-transformer integration can capture subtle structural details in intricate textures, but they are limited by turbidity, lighting variation, and visual unclarity among classes according to morphological similarity.

Overall, the figure shows that, in difficult underwater settings, the suggested framework can detect multiple classes and objects. The system combines transformer-based contextual reasoning with CNN-based local texture extraction to achieve strong recognition ability over a variety of coral morphologies. This demonstrates that the proposed model is appropriate for monitoring ecosystems and evaluating the health of reefs.

To demonstrate the robustness of the proposed model in detecting coral morphology from underwater coral imagery we have incorporated zoomed-in view to better illustrate the object detection results of a single coral image. Figure 5 presents a representative example illustrating the model's performance for a zoomed in image. The figure shows the visual comparison of the ground truth and the predicted output. The Fig. 5a displays the original underwater image containing arborescent coral under natural lighting and visibility conditions. The coral image in Fig. 5b shows final detection result generated by the proposed model. The bounding boxes demonstrate the precise detection with correct classification and for the arborescent coral with high confidence score. The visualization confirms the model's improved spatial discrimination and robustness in complex maritime environments. The presented results confirm the advantages of the proposed CNN-Transformer fusion framework in capturing fine-grained coral structures and complex morphological patterns

Thus, the figure shows the robust performance of the proposed model in underwater conditions for fine-grained automated coral structural identification. This visual evidence shows the potential of the proposed model for reliable coral identification and marine ecosystem monitoring.

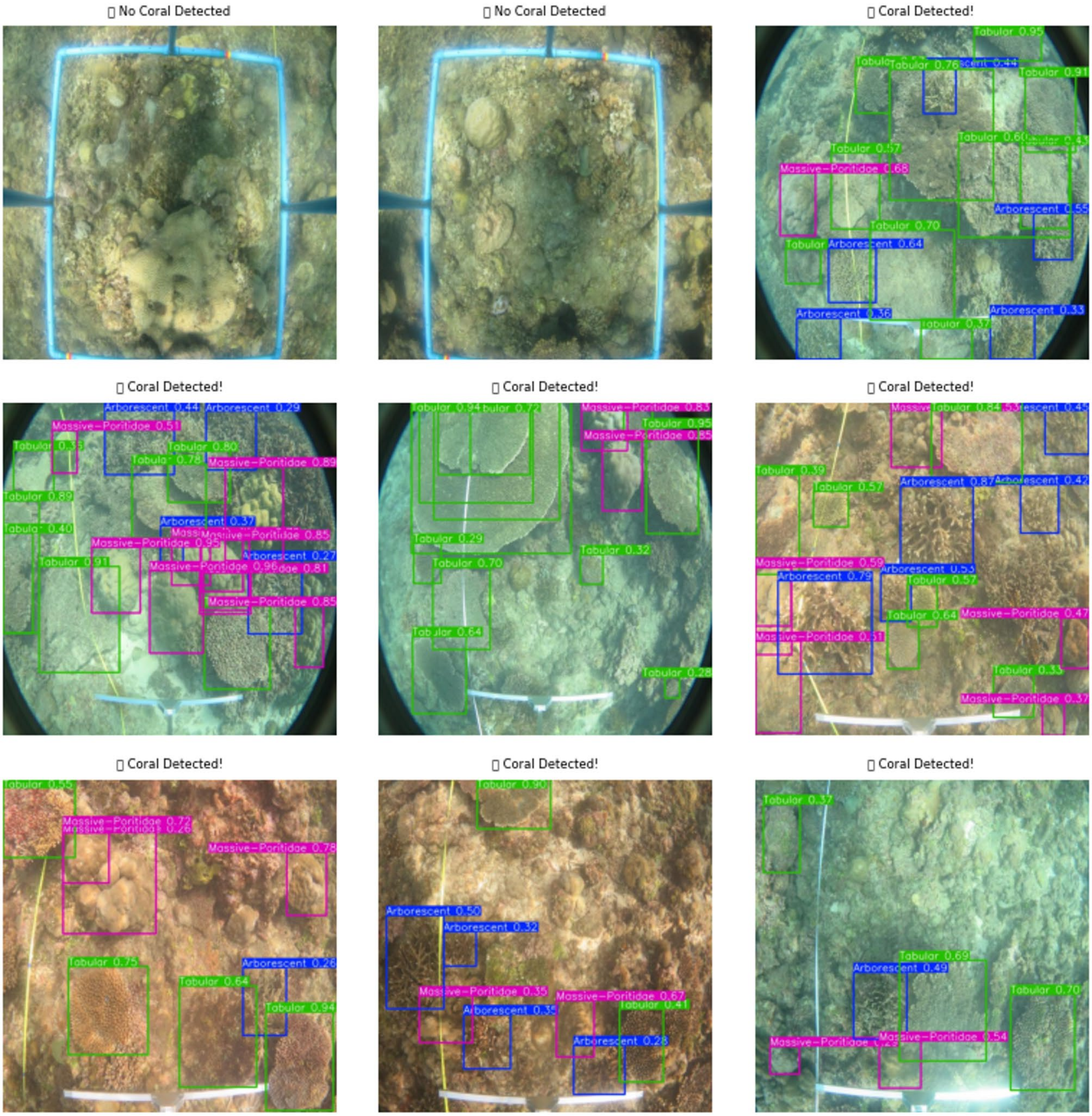

Fig. 4. Detection results of YOLOv12 on underwater imagery.

Key evaluation metrics during training for the proposed YOLOv12 with CNN-transformer encoder fusion for coral morphology detection are shown in Fig. 6. The Precision, Recall, mAP@0.5, and mAP@0.5:0.95 on the validation dataset are shown. All the curves have an upward inclination as the number of iterations increase, indicating gradual gains in recognition ability while the network adjusts its parameters. The sharp rise in precision during the initial phase before it stabilizes indicates that the model quickly learns to reduce false positives in distinguishing coral morphologies from background structures. Similarly, recall also shows an analogous curve, demonstrating the system's growing ability to detect corals in a variety of underwater settings. The mAP@0.5 measure attains higher values promptly during training. This illustrates that the model can execute bounding box localization well at a moderate IoU threshold.

In contrast, the stringent mAP@0.5:0.95 curve ascends smoothly, emphasizing how challenging it is to localize accurately in underwater images where corals are frequently overlapping, comprise asymmetrical forms, and somewhat obscured. The transformer encoder module most effectively presents features of global context, while the CNN encoder maintains the finest performance in embodying subtle local coral textures pertaining to corals. The combination leads to precise and generalizable recognition of coral morphology across various underwater conditions.

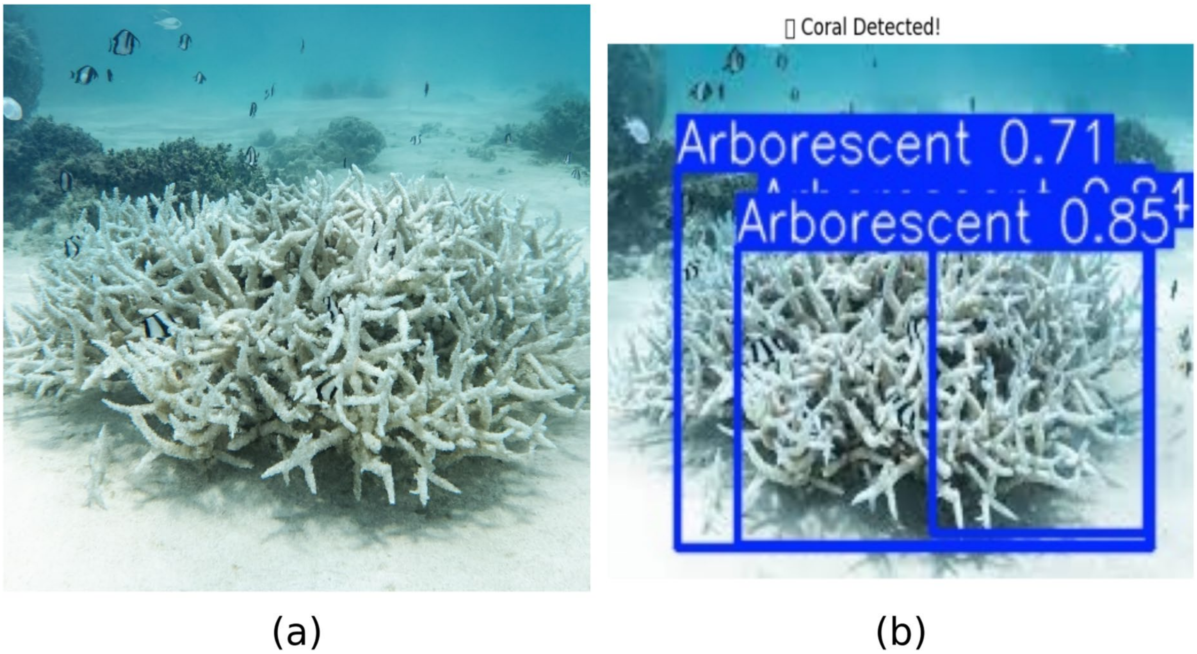

Fig. 5. Detection results of the proposed model for zoomed-in view on underwater image.

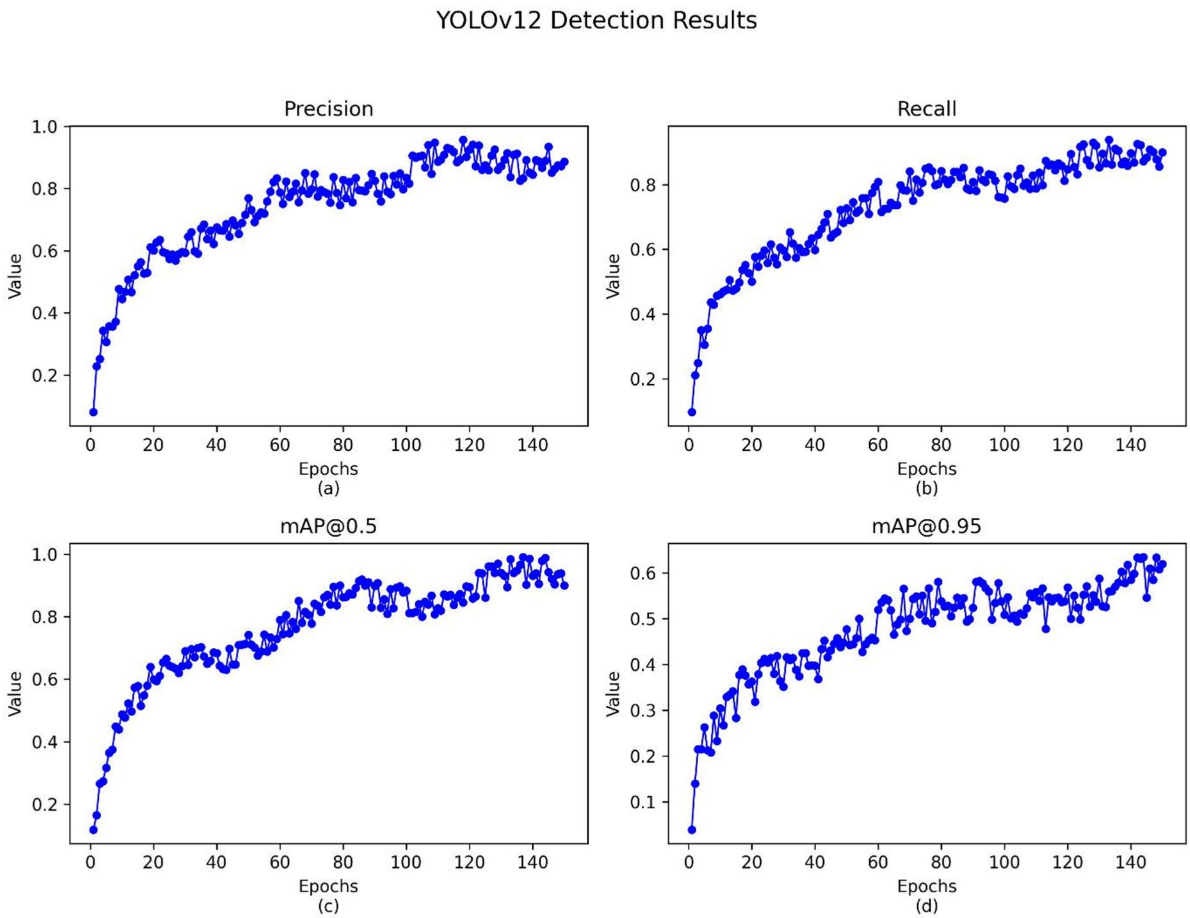

Fig. 6. Performance of various metrics.

Thus, the concurrent enhancement across all metrics justifies the efficacy of the CNN-transformer integration approach in elevating both the accuracy of detection and precision of localization. The steady ascending progression devoid of significant oscillations point towards steady training dynamics, which underlines that the suggested model generalizes well towards the intricacies inherent in identification of underwater coral morphology.

The Precision-Recall (PR) curves in Fig. 7 describe the performance of the proposed YOLOv12 method, incorporating a fusion of a CNN and a transformer encoder on the three morphologies of the corals, arborescent, tabular, and massive. The PR curve depicts the precision-recall trade-off and is useful for unbalanced datasets like underwater coral images. The Area Under the PR Curve (AUC-PR) values reported are high at 0.88 for arborescent, 0.90 for massive, and 0.91 for tabular corals. The higher values of AUC-PR reflect that the method predicts high precision and recall values at different thresholds. For the three categories of corals, the tabular corals report the maximum value of AUC-PR, which confirms that the tabular structure of the corals is easy to recognize and differentiate. The second closest is massive corals, indicating a good positive classification while suggesting some overlap between classes. Arborescent corals show marginally reduced AUC-PR value, mostly due to the complex, fine, branching structure and a high possibility of these branches getting hidden, thus making detection a difficult job. The combination of CNN feature learning capabilities with a strong global understanding using a transformer helps greatly in minimizing both false-positive and false-negative rates in intricate underwater settings. Analysing the PR-curve verifies that the combination of CNN and transformer performs a remarkable job on detection tasks pertaining to different types of corals. The attained high AUC-PR values reflect that the system retains high precision and recall over a broad range of thresholds, supporting strong detection under complicated underwater environments. Tabular corals have the best AUC-PR (0.91) of all classes, indicating that their unique morphological characteristics are easier to recognize and differentiate. Integrating CNN-based local feature learning with transformer-based global contextual understanding efficiently balances precision and recall, minimizing false positives and false negatives in rich underwater settings. The PR curve analysis generally verifies that the suggested CNN-transformer fusion strongly improves detection ability for various coral morphologies.

Figure 8 shows the normalized confusion matrix for the classification of the coral morphologies derived through the proposed YOLOv12 model integrated with CNN and transformer encoder. The matrix represents the predictive capacity of the model on the three classes of morphologies, where the rows and columns represent the true classes and the predicted classes, respectively. The values are normalized to facilitate the interpretation. Correct classifications are indicated by the diagonal elements, where the model makes predictions that are 96% accurate for tabular, 94% accurate for massive, and 93% accurate for arborescent morphology class. Robust discrimination ability is depicted by these results, which suggests that the combined local (CNN) and global (Transformer) features successfully record the textural and structural characteristics of the corals. Off-diagonal cells represent the misclassification output. For the tabular morphology class, there is approximately 4% misclassification. Also, there is approximately 6% misclassification of the massive morphology type to either the arborescent or the tabular morphology class. The highest level of misclassification is shown by the

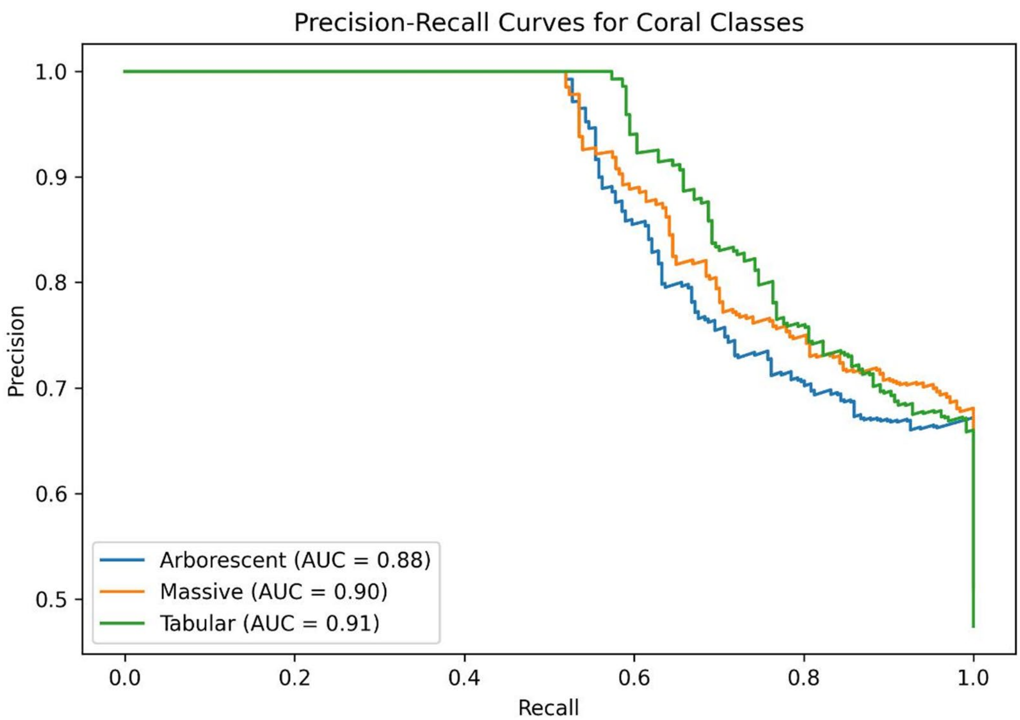

Fig. 7. Precision-Recall curves for coral classes.

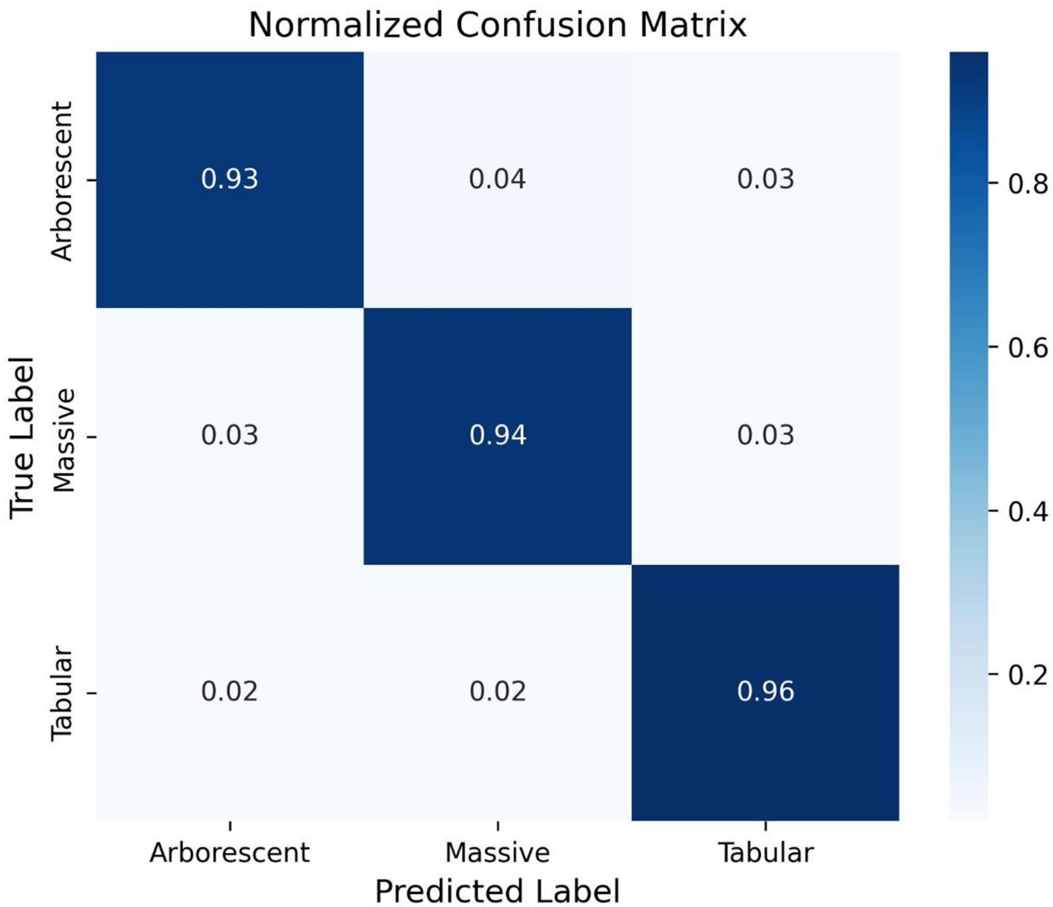

Fig. 8. Normalized confusion matrix.

arborescent morphology class of corals. The percentage of correct predictions is 93%, while those misclassified are approximately 4% massive and 3% tabular. Overlapping visual features are exhibited by the arborescent morphology class with other classes. This is because of some occlusion, low contrast in underwater coral images, or basic morphological similarity across classes.

In conclusion, the confusion matrix points out both the strengths and the issues of the suggested framework. The model is able to differentiate well between different forms of corals, but the nuances in the boundaries between the forms sometimes make it hard to distinguish. This serves to illustrate the importance of the CNN and transformer model integration in YOLOv12, since the CNN part contributes the detail in the textures, while the transformer part contributes the understanding of the context.

Figure 9 shows how the identified coral morphologies are spread out in space according to the model's classification. The dots show where corals were found, and the colours show what kind of coral they are: green for arborescent, orange for tabular, and red for massive. The pattern of the scatter reflects the ability of the model to distinguish different shapes over a variety of underwater settings. One can notice a group of green points, indicating that arborescent corals, with their branching forms, are recognized consistently. Tabular corals appear more equitably dispersed, matching their flat, plate-like growth caught in the images. Massive corals are marked in red colour, which are usually concise and dome-shaped. This local clustering of massive corals suggests that the model is reliably sensitive to this morphology as well. Overall, this visualization illustrates strong multi-class detection efficacy and highlights the capacity of the model to generalize across varied spatial patterns. The balanced class representation also highlights the efficacy of combining spatial convolutional features with transformer-inspired global context modelling for fine-grained coral morphology classification in complex underwater settings.

Comparative analysis

Figure 10 shows a comparison of how well various coral morphologies (arborescent, tabular, and massive) were able to be detected on several tested architectures, employing mAP@50 as a measure. This graph shows the performance of each model. It also shows that the proposed YOLOv12-CT model does better than all the benchmark models based on the evaluation metric. This is especially clear with tabular corals, where YOLOv12-CT has a mAP@50 value of over 0.90, which is much higher than that of YOLOv12 and shows big improvements over all models.

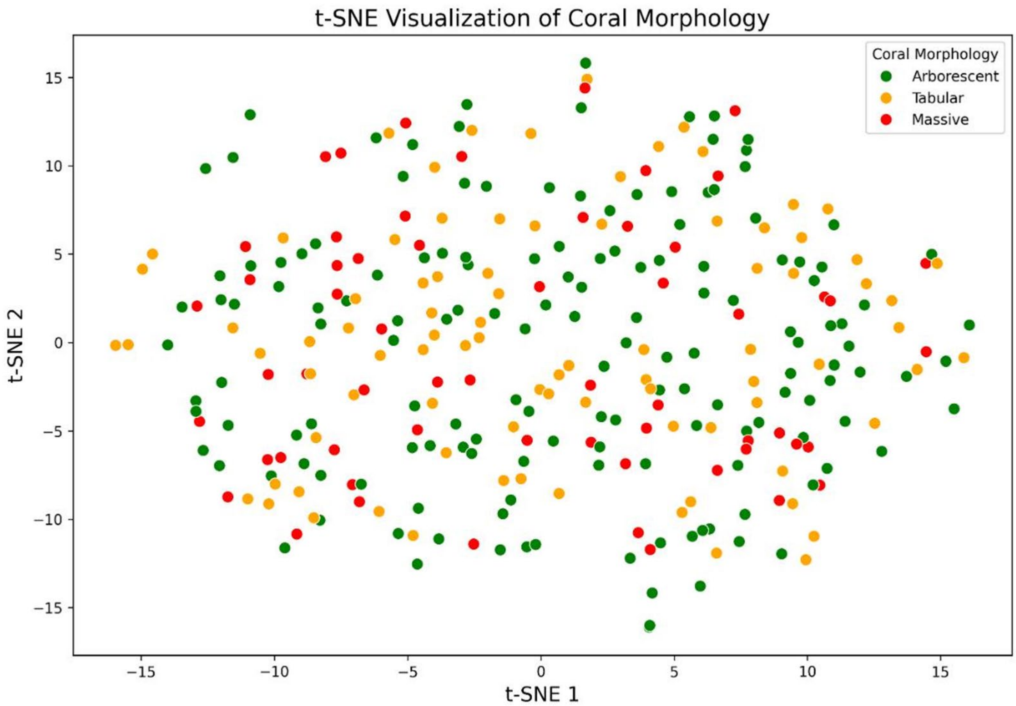

Fig. 9. Spatial distribution of coral morphology detections using the proposed method. Points represent detected corals, colour-coded by morphology: arborescent (green), tabular (orange), and massive (red).

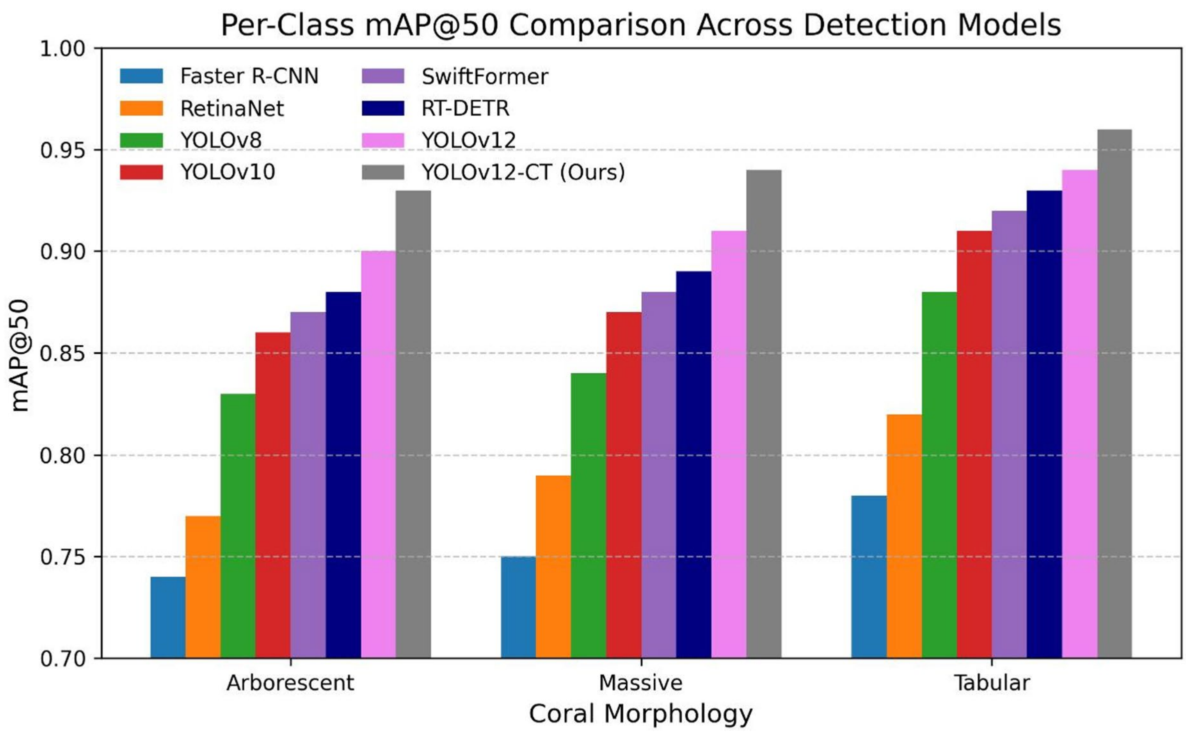

Fig. 10. Per-class detection performance.

The combination of a CNN and a transformer encoder in YOLOv12 is much better than sole YOLOv12. This combination brings noteworthy improvements in performance, emphasizing the importance of integrating convolutional feature extraction and global attention to facilitate the study of the morphology of corals at a finer level. This is very important, especially in the underwater scenes where the conditions of identifying the reef structures are difficult. Transformer-enhanced models like SwiftFormer and RT-DETR perform better than solely CNN-based methods on all counts. These methods display significant improvements for arborescent and massive coral classes. This shows how global context modelling and better representation of complex spatial relations can be helpful. Across all classes, RT-DETR has a higher mAP@50 than SwiftFormer. The performance of the conventional CNN detectors, namely Faster R-CNN and the RetinaNet detector, remains lower for each class of coral. The reduction in detection accuracy, specially for arborescent coral class, occurs due to their inadequate capability to detect intricate structural features and spatial relationships at a longer range. YOLOv8 can be considered an improvement compared to these previous approaches. YOLOv10 improves the performance for each class, specifically for the branching type of coral morphology, revealing better localization and feature aggregation abilities.

That higher value of mAP@50 indicates that the flat and plate-like morphology of tabular corals is more salient against the backdrop of the seafloor and helps the model to detect them more easily. Arborescent corals have somewhat lower mAP@50 values because of their more delicate and branching patterns, similar to those of a tree, and the often-present overlaps visible in the images of reefs. Massive corals have intermediate values because of their large and dome-like morphologies that sometimes make them blend in with the surrounding reefs. Nevertheless, the model's performance has been relatively stable across different morphologies, indicating that it is capable of generalizing to structurally diverse corals.

The comparison draws attention to the fact that the newly suggested YOLOv12-CT not only outperforms the past YOLO models, but it outperforms the transformer-based models as well as the conventional CNN-based models on the basis of detection performance. The study not only indicates the well-balanced detection ability of the proposed model to detect coral morphologies in the challenging underwater conditions, but it is also practical for fine-grained coral morphology detection. On the whole, the bar chart proves its efficacy for the integration of the CNN and the transformer encoder in the YOLOv12 model, suggesting that the newly suggested YOLOv12-CT achieves excellent detection performance for the coral morphologies in underwater images.

Figure 11 compares several deep learning models to detect coral morphologies in underwater images. As a result of this comparison, we are able to make a balanced assessment of the suitability of each algorithm for underwater coral reef morphology detection tasks based on both detection effectiveness (accuracy, precision, recall, mAP50, and mAP50-95) and computational efficiency (inference time, parameter count, and FLOPs).

Both two-stage and early single-stage detectors, particularly Faster R-CNN and RetinaNet, demonstrate relatively lower efficiency in all object detection metrics. Faster R-CNN attains the lowest mAP50-95 at 0.43, highlighting its confined ability to produce precise localization of complex and fine-grained coral morphologies under variable underwater conditions. RetinaNet attains modest improvements due to its feature pyramid network and focal loss; nonetheless, its recall and localization accuracy are bounded at 0.72 and 0.74 when dealing with irregular coral morphologies and small-scale features. RetinaNet shows mAP50-95 value of 0.45. Such findings show that coral morphologies have a lot of variation within the same class, which solely CNN-based models without advanced feature aggregation ability cannot handle.

YOLOv8 and YOLOv10 show a noteworthy improvement in detection performance, attaining substantial increase in precision, recall, and mAP scores. YOLOv10 is better than all the previous YOLO series versions. Its mAP50-95 score of 0.47 shows that it has better localization and is robust in crowded coral scenes. Transformer-based models make accuracy even better by the means of global contextual understanding. SwiftFormer and RT-DETR performs better than CNN-only YOLO variants in mAP50-95 by attaining scores of 0.50 and 0.52,

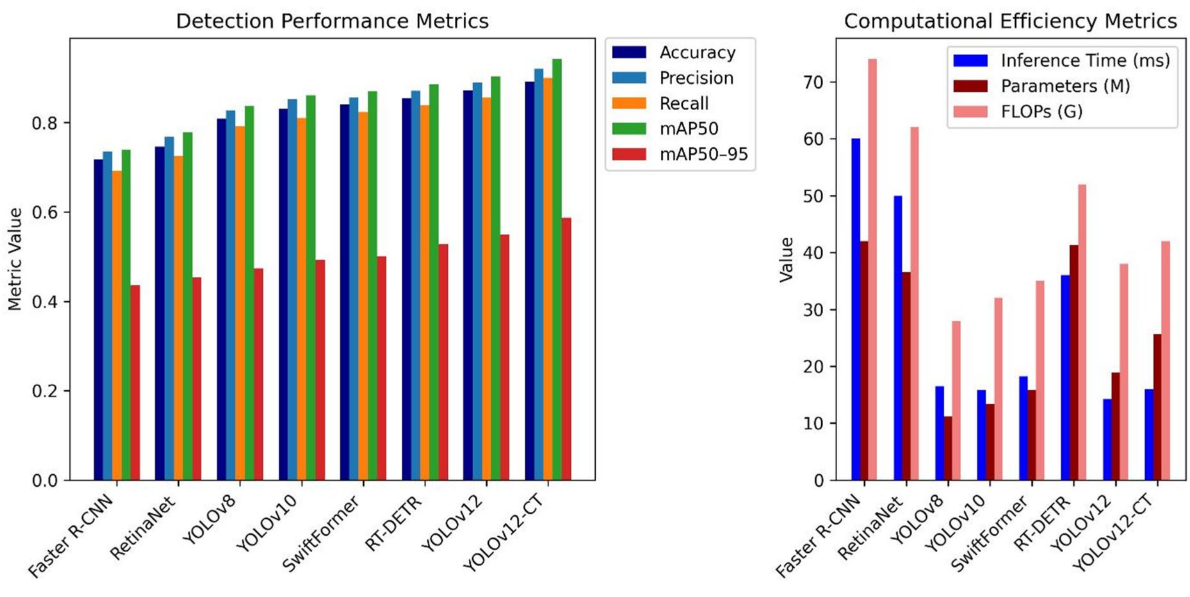

Fig. 11. Performance comparison of various models.

respectively. RT-DETR had a higher accuracy and recall of 0.85 and 0.83, respectively, compared to SwiftFormer, which had an accuracy and recall of 0.84 and 0.82, respectively. This shows that they can handle occlusions, branching structures, and asymmetrical coral boundaries better.

YOLOv12 shows further improvements in performance and reaches at accuracy 0.87, precision, 0.88, recall, 0.85 and mAP50-95 0.55. The best one is the hybrid version of YOLOv12, namely, YOLOv12-CT (CNN + Transformer), which yields the superior values for all metrics. It attains an accuracy 0.89, precision 0.92, recall 0.90 and mAP50 0.94. Its mAP50-95 value of 0.58 is especially good, making it the most robust over multiple IoU thresholds. Combining transformer encoder and CNN in YOLOv12 to extract local feature and long range relations is useful for detecting detailed coral morphology.

By comparing inference time, parameter count, and FLOPs, the right subplot evaluates each model's computational efficacy. These metrics describe the resource requirements of the evaluated models. These metrics help in evaluating the suitability of the models for real-time deployment to detect coral morphologies. The subplot illustrates the longer inference times of 60 ms and 50 ms, respectively, for two-stage detectors like Faster R-CNN and RetinaNet. On the other hand, YOLOv8, YOLOv10, and YOLOv12 have much lower inference latency of 16.5 ms, 15.8 ms, and 14.2 ms, respectively. Even though RT-DETR and YOLOv12-CT use transformer modules, their inference times are still reasonable. RT-DETR has higher inference time than YOLOv12 and its variant YOLOv12-CT. The minimum inference time is shown by YOLOv12 14.2 ms. The proposed YOLOv12-CT exhibit an inference latency of 16ms. Parameter count bar graphs illustrate that Faster R-CNN, RT-DETR, and RetinaNet use maximum number of parameters, 42 M, 41.3 M, and 36.5 M, respectively. This is because they have intricate feature processing stages. YOLO-based models and SwiftFormer use lesser parameters. The SwiftFormer has a parameter count of 15.8 M. YOLOv8, YOLOv10 and YOLOv12 use a parameter count of 11.2 M, 13.4 M and 18.9 M respectively. The proposed YOLOv12-CT uses a parameter count of 25.6 M. Due to the addition of transformer encoder, the suggested model shows a slight increase in parameters, but it is still significantly lighter than conventional two-stage or full transformer detectors.

High FLOPS are illustrated by models such as Faster R-CNN, RetinaNet and RT-DETR, 74 G, 62 G, and 52G respectively. On the other hand, YOLO-family models show markedly lower FLOPs, underscoring their efficiency-oriented design. YOLOv8, YOLOv10 and YOLOv12 display 28 G, 32 G, and 38 G FLOPS respectively. The proposed YOLOv12-CT model depicts 42 G FLOPs. Transformer-based models have more FLOPs than YOLO-based models, but they do not scale in proportion to inference time. This highlights the fact that runtime performance is not entirely determined by FLOPs alone.

Even though YOLOv12-CT costs a little more to run than YOLOv12, its higher accuracy makes it good for tasks that require very precise ecological monitoring. The rise in inference time of the suggested model is modest in spite of the fusion with transformer encoder and CNN. This shows how the suggested hybrid model strikes a good balance by improving detection performance without requiring too much additional processing power. The study of computational efficiency shows that the combined YOLOv12-CT framework can be deployed in real-world underwater monitoring systems to detect coral morphologies, where real-time performance is frequently vital.

In short, the comparative study demonstrates that combining CNN and transformer encoder in YOLOv12 greatly improves the accuracy of detecting coral morphologies, predominantly when the conditions for underwater imaging are difficult, while keeping the computational efficiency balanced. This confirms that the suggested method is a good way to monitor coral reefs on a large scale.

Ablation experiments

To verify the efficacy of the suggested CNN-transformer encoder fusion in YOLOv12 for coral morphology detection, we executed an extensive series of ablation experiments. The influence of each component is isolated in the experiments to detect the effect on performance. The same dataset, training protocols, input resolutions and hyperparameters were used to guarantee an equitable comparison. All models were trained for 150 epochs with the same optimizer, learning rate schedule, batch size, and data augmentation strategy.

Table 1 presents the results of the ablation study, which examines the collaborative and individual contributions of the CNN enhancement and transformer module to enhance the YOLOv12 architecture. The reference point is a baseline YOLOv12 model. It has a precision of 0.838 and a recall of 0.832, and its mAP@50 and mAP@50–95 scores are 0.876 and 0.513, respectively. These outcomes show how difficult it is to detect coral morphology, as there is a lot of variation within classes, and distinguishing backgrounds is particularly difficult in underwater images.

When CNN-based feature refinement is added, the model's performance on all metrics improves moderately in a steady manner. The mAP@50 score goes up to 0.899, the precision score goes up to 0.859, and the recall score goes up to 0.858. This enhancement shows that improved local feature extraction makes the model better at capturing the small details of coral formations' textures and structures.

Model variant	Precision	Recall	mAP@50	mAP@50–95
YOLOv12 (Baseline)	83.8	83.2	87.6	51.3
YOLOv12 + CNN	85.9	85.8	89.9	53.2
YOLOv12 + Transformer Encoder	88.6	88.1	91.8	55.1
Proposed YOLOv12-CT	**91.4**	**90.1**	**94.2**	**58.5**

Table 1. Results of the ablation study for the proposed method.

Model variant	Precision	Recall	mAP@50	mAP@50–95
Backbone (Early fusion)	89.3	88.2	92.6	56.5
Neck (Intermediate fusion)	**91.4**	**90.1**	**94.2**	**58.5**
Head (Late fusion)	88.1	87.4	91.4	55.3

Table 2. Ablation study to show the impact of fusion location in the proposed method.

Adding the transformer encoder makes the enhancements in efficacy even bigger. The precision and recall scores rise up to 0.886 and 0.881, respectively, and the mAP@50 and mAP@50–95 scores also surge up to 0.918 and 0.551, respectively. These findings highlight the efficacy of the transfer encoder module for modelling long-range spatial relationships, which are specifically useful for finding coral morphologies that are uneven and spatially distributed.

The suggested YOLOv12-CT method performs exceptionally well when both CNN refinement and the transformer encoder are fully integrated. The model has a precision of 0.914 and a recall of 0.901. The mAP@0.5 and mAP@50–95 scores are 0.942 and 0.585, respectively. The steady progress across all ablated versions validates that CNN-based local feature learning and transformer-based global contextual modelling yield synergistic advantages, specifically for intricate coral morphologies characterized by uneven boundaries.

We subsequently analysed how incorporating the CNN and transformer encoder at different stages in YOLOv12 affected the model's performance. Table 2 shows the results of the ablation study on how the location of the fusion affects the performance of the proposed architecture. Fusion at the neck stage works best because it keeps fine spatial details while allowing for global contextual reasoning. This is important for finding both large coral colonies and fine-grained branching structures. These results confirm the architectural design decisions and illustrate the essential role of each component in attaining reliable and precise coral morphology detection.

Limitations and deployment challenges

This research possesses numerous limitations. The RF100 dataset consist of underwater coral images taken in low light and from different camera angles. Many of these images were taken under regulated visibility conditions, and as a result, the dataset does not contain full conditions that are common in usual underwater domain such as turbidity, colour attenuation, and harsh interference. Also, even though the dataset has a lot of different coral morphologies, there is still a class imbalance within the dataset. For example, tabular and massive corals are excessively prevalent in relation to lesser common variants. This class imbalance is likely to result in the model making incorrect predictions.

Furthermore, the integration of CNN and transformer encoders enhanced local feature extraction and global spatial attention within the dataset; however, it may not ensure consistent efficiency on unobserved reef systems or under diverse sensor types. The generalizability of the model might be even more limited by variances in water depth, light spectra, and camera optics that RF100 dataset doesn't show. To use the model broadly use, cross-domain adaptation and refinement on images from specific regions are needed.

Moreover, though the transformer encoder made it easier to extract global context, but it also made the model bigger and more expensive to run, which could make it harder to use in real time on low-powered embedded underwater platforms. Using lightweight transformer modules, model pruning, or knowledge distillation to optimize the fusion model could help find a good balance between accuracy and efficacy for deploying on edge devices. Underwater illumination and chromatic distortion are still hard to deal with; while augmentation helps, the real world is more variable. Lastly, the assessment was limited to static images; incorporating video-based detection or three-dimensional structures would enhance its real-world applicability.

Conclusion and future work

This paper introduced an innovative method to identify coral morphologies from underwater images, utilizing YOLOv12 model augmented with CNN-transformer encoder fusion. The suggested approach solves important problems to monitor coral reefs automatically. The model combines the ability of CNN to extract the local features and transformer encoder to model global context. The projected method performed effectively in difficult underwater settings, demonstrating its efficiency. Research findings show noteworthy gains in all assessment metrics. The suggested method achieves 0.94 mAP@50, which is a considerable progress from current approaches. The integrated framework is especially good at dealing with the different shapes of coral structures. It consistently improves performance on branching, massive, and tabular coral classes. The results indicate that the suggested method is appropriate for implementation to monitor coral morphologies automatically where environmental settings are extremely erratic and frequently suboptimal. The investigation of the computational efficiency shows that the suggested approach ensures practical relevance for real-time applications. It shows an inference speed of 17 milliseconds and significant advances in accuracy. The CNN-transformer fusion in YOLOv12 method makes a big difference in performance, so the small increase in computing power is worth it.

Future endeavors will concentrate on various important aspects: incorporating temporal data by using video sequence analysis so that occlusion handling can be improved, specialized attention mechanisms can be developed for underwater environments, coral species can be classified with much detail. For large-scale reef monitoring, the model can be used with AUV platforms. Adding other types of data, like sonar, hyperspectral imaging, or 3D point clouds, can assist with occluded corals and make it easier to tell the difference between coral species that look similar in shape.

Thus, the suggested CNN-transformer integration method is a big step forward to monitor coral reefs in an automated way. It gives marine biologists and conservationists better tools for quickly and accurately assessing coral ecosystems. The method works better and is more useful in real life, which makes it a good choice for use in operational marine monitoring systems.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

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Declarations

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